Intravital 2-photon imaging reveals distinct morphology and infiltrative properties of glioblastoma-associated macrophages

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Characterized by a dismal survival rate and limited response to therapy, glioblastoma (GBM) remains one of the most aggressive human malignancies. Recent studies of the role of tumor-associated macrophages (TAMs) in the progression of GBMs have demonstrated that TAMs are significant contributors to tumor growth, invasion, and therapeutic resistance. TAMs, which include brain-resident microglia and circulating bone marrow derived-monocytes (BMDMs), are typically grouped together in histopathological and molecular analyses due to the lack of reliable markers of distinction. To develop more effective therapies aimed at specific TAM populations, we must first understand how these cells differ both morphologically and behaviorally. Furthermore, we must develop a deeper understanding of the mechanisms encouraging their infiltration and how these mechanisms can be therapeutically exploited. In this study, we combined immunocompetent lineage tracing mouse models of GBM with high-resolution open-skull 2-photon microscopy to investigate the phenotypical and functional characteristics of TAMs. We demonstrate that TAMs are composed of 2 morphologically distinct cell types that have differential migratory propensities. We show that BMDMs are smaller, minimally branched cells that are highly migratory compared with microglia, which are larger, highly branched stationary cells. In addition, 2 populations of monocytic macrophages were observed that differed in terms of CX3CR1 expression and migratory capacity. Finally, we demonstrate the efficacy of anti-vascular endothelial growth factor A blockade for prohibiting TAM infiltration, especially against BMDMs. Taken together, our data clearly define characteristics of individual TAM populations and suggest that combination therapy with antivascular and antichemotaxis therapy may be an attractive option for treating these tumors.

glioblastoma | macrophage | two-photon | monocyte | microglia

Despite aggressive therapeutic intervention, glioblastoma (GBM; World Health Organization grade IV) remains one of the deadliest human cancers, with a median survival of approximately 15 mo (1). Current therapeutic regimens target properties of neoplastic cells but fail to target prevalent nonneoplastic cell populations of the tumor, including tumor-associated macrophages (TAMs). TAMs symbiotically interact with tumor cells through the secretion of growth factors, matrix metalloproteinases, and other factors that promote tumor cell invasion and progression, thus rendering them attractive targets for therapeutic intervention (2, 3). We have previously demonstrated that TAMs, which consist of resident-brain microglia and bone-marrow derived monocytes (BMDMs), constitute up to 40% of the total tumor mass in both human and mouse GBMs (2, 4). Furthermore, we have demonstrated in a PDGF-B–driven genetically engineered mouse model (GEMM) of GBM that >80% of these cells are BMDMs that infiltrate from the vasculature (5). A recent elegant study using single-cell profiling of human GBMs confirmed our data and showed that monocytes significantly infiltrate pretreatment human gliomas to a degree that varies by glioma subtype and tumor compartment (6). BMDMs do not universally conform to the phenotype of microglia, but preferentially express immunosuppressive cytokines and show an altered metabolism. These findings, together with ours, argue against traditional therapeutic strategies that target TAMs indiscriminately and instead favor strategies that specifically target immunosuppressive BMDMs.

Monocytes develop from hematopoietic stem cell progenitors and remain in the intact vasculature of the brain, with their infiltration into surrounding tissue hindered by the blood brain barrier (2). However, during GBM tumor progression, increased expression of monocyte chemoattractant proteins (MCPs) (7) in combination with the disruption in blood brain barrier integrity results in infiltration of inflammatory monocytes into the perivascular niche (8). Once inside the tissue, infiltrating monocytes can then differentiate into macrophages and exert protumor effects (9). We have previously demonstrated that a genetic decrease in the CCL2/CCR2 axis in GBMs reduces monocyte infiltration but is insufficient to completely ablate it, suggesting that other mechanisms are responsible for the recruitment of these cells as well (5).

Significance

Therapeutic strategies targeting glioblastoma (GBM) tumor-associated macrophages (TAMs) have shown limited efficacy. One reason for this limited effectiveness is our poor understanding of the fundamental characteristics driving the behavior of these cells. Here we report a combination of immunocompetent lineage tracing mouse models of GBM with an open skull window for in vivo observation of TAM constituents via 2-photon microscopy. We show that TAMs are composed of 2 morphologically and behaviorally distinct cell types: brain-resident microglia and bone marrow-derived macrophages (BMDMs). After treatment with anti-vascular endothelial growth factor A antibody, we observed significant reductions in BMDM infiltration and morphological responses. These studies provide a platform for monitoring the multidimensional responses of specific cell populations to therapeutic intervention in real time.

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Unlike BMDMs, microglia are always present in the central nervous system and are homogeneously dispersed throughout the brain. These long-lived cells, which develop from embryonic yolk sac progenitor cells during embryogenesis, are responsible for the clearing of pathogens and maintenance of synaptic processes (10, 11). In their activated state, microglia produce proinflammatory cytokines and reactive oxygen species, assume an amoeboid morphology, and become histologically indistinguishable from infiltrating BMDMs (12, 13). In contrast to BMDMs, which are found in tumor bulk and infiltrating tumor margins, microglia are sparsely distributed throughout the tumor and accumulate at the periphery. Analysis of freshly dissociated patient GBM samples for BMDMs and resident microglia revealed significant infiltration of BMDMs across the GBM subtypes. Only the mesenchymal subtype of GBM showed increased microglia infiltration compared with BMDMs (14).

Two-photon microscopy has previously been used for the longitudinal analysis of GBM growth kinetics, vascular dynamics, and immune cell infiltration (15–18). However, the contributions of individual TAM populations have largely been ignored, because microglia and circulating monocytes are typically grouped together and treated as a single collective entity. Although we have a firm understanding of TAMs in the context of tumor growth promotion and immune suppression, the majority of published histopathological studies are limited to static snapshots of the disease over time. If we seek to identify therapeutic points of intervention, we must understand how these cell types interact with their microenvironment in real time. Furthermore, the development of cell type-specific targeted therapies will be possible not only by studying the TAM population as a whole, but also by identifying biological differences between the two aforementioned subtypes.

In the present study, we performed live in vivo 2-photon microscopy in an immunocompetent GEMM of GBM together with myeloid cell-specific reporter mice to lineage-trace microglia and BMDMs. By taking advantage of reciprocal bone marrow chimeras between *Cx3cr1*$^{GFP/WT}$ and wild-type (WT) mice, we demonstrated the feasibility of using an open-skull window for the imaging of TAM subpopulations in tumor tissue. We observed that tumor microglia and infiltrating BMDMs are morphologically distinct cell populations with differing migratory capacities. We further showed that anti-vascular endothelial growth factor A (VEGFA antibody) treatment significantly decreases infiltration and induces a morphological switch in BMDMs to resemble differentiated macrophages. This study also demonstrates that blood-brain barrier (BBB) integrity is not the sole driver of monocyte infiltration and provides a rationale for combining antiangiogenic and antichemotaxis (targeting members of the MCP family) therapies to block monocyte infiltration.

Results

Two-Photon Imaging Permits Direct, Longitudinal Observation of TAMs In Vivo. Advantages of 2-photon microscopy over traditional confocal microscopy include reduced autofluorescence and photobleaching effects, increased imaging depth, and minimal photodamage to surrounding brain tissue (19). Consequently, we used 2-photon microscopy for the in vivo analysis of individual TAM populations. To minimize breathing artifacts during imaging, a custom acrylic adapter that attaches to the cranium after skull window placement was manufactured in-house. This adapter attaches to a stainless-steel stage that holds the mouse in place throughout image acquisition (Fig. 1*A*). Superficial tumors are easily visualized through the skull window and can be distinguished from adjacent normal brain tissue by eye or at low magnification with a stereoscope (Fig. 1*B* and *SI Appendix*, Fig. S1). To avoid visual artifacts induced from skull window placement, such as glial scarring and leukocyte accumulation, all image stacks were analyzed at a depth of 50 μm or deeper from the pial surface, typically between 100 and 500 μm (Fig. 1 *C* and *D*). High-resolution image stacks with approximate dimensions of 400 μm × 400 μm × 400 μm were typically acquired and analyzed.

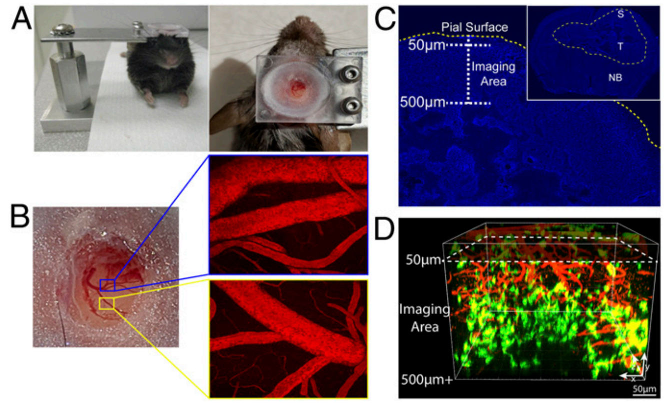

Fig. 1. The 2-photon imaging setup and acquisition procedures. (*A*) A custom acrylic adapter attaches the mouse to a steel stage to stabilize the imaging plane and reduce movement artifacts. (*B*) Superficial tumors and vascular landmarks can be observed by eye and at low magnification. (*C*) All image analysis was performed at a depth of >50 μm below the pial surface and down to ∼500 μm deep. Blue indicates DAPI stain. S, surface; T, tumor; NB, normal brain. (*D*) High-resolution Z stacks were imported into Imaris for image processing and analysis. TRITC, vessels; GFP, TAMs.

Although not performed in this study, this methodology also allows for the visualization of other cell types and cellular components tagged with fluorescent molecules, such as tumor cells, lymphocytes, microtubules, and others (18, 20, 21).

Visualization of TAM Populations in Chimeric Mice. Using a *Cx3cr1*$^{GFP/WT}$;*Ccr2*$^{RFP/WT}$ mouse model, we previously demonstrated that infiltrating peripheral monocytes show decreased CCR2-RFP expression as they mature in the GBM microenvironment. Since we are interested in quantifying both newly infiltrating monocytes and differentiated macrophages, this model is not optimal for our studies, as we would inevitably miss a large population of cells as they mature. To remedy this, we used reciprocal bone marrow chimera mouse models with 1 allele of the *Cx3cr1* gene replaced with GFP (Fig. 2*A*). These models allow us to discern differences between infiltrating BMDMs and brain-resident microglia. CX3CR1 is the sole receptor for the chemokine CX3CL1 (fractalkine), which is widely expressed throughout the central nervous system. Replacement of 1 allele with GFP (*Cx3cr1*$^{GFP/WT}$) has been demonstrated to have no effect on the migration capacity or infiltration of TAMs (22, 23).

To generate mice with GFP-expressing BMDMs and WT microglia (i.e., no GFP expression in microglia), bone marrow from *Cx3cr1*$^{GFP/WT}$ mice was mixed in a 1:1 mixture with bone marrow from *Cx3cr1*$^{WT/WT}$ mice. This mixture was then injected into whole-body–irradiated mice to reconstitute the bone marrow with 50% GFP-expressing BMDMs and 50% WT BMDMs (*SI Appendix*, Fig. S2). This scheme allows for single-cell resolution of infiltrating BMDMs, whereas a nondiluted GFP signal poses difficulties for resolving individual cells due to the high density of these cells in the tumor. Non–tumor-bearing mice had few to no GFP-expressing BMDMs in the brain parenchyma, as these cells do not readily cross the BBB (Fig. 2*B* and *SI Appendix*, Fig. S3 *A–D*); however, imaging of tumor-bearing mice revealed a significant infiltration of GFP-expressing BMDMs (Fig. 2 *B–D*). These cells were seen to widely populate tumor bulk, perinecrotic areas, and infiltrative tumor margins, while their presence was absent in adjacent nontumor tissue (*SI Appendix*, Fig. S4*B*).

To generate mice with WT BMDMs and GFP-expressing microglia, donor bone marrow isolated from *Cx3cr1*$^{WT/WT}$ mice was injected into whole-body–irradiated *Cx3cr1*$^{GFP/WT}$ recipients. Microglia populate the brain during embryogenesis and are found consistently dispersed throughout the tissue (24). In tumor-bearing brains, microglia are sparse in tumor bulk but often accumulate in

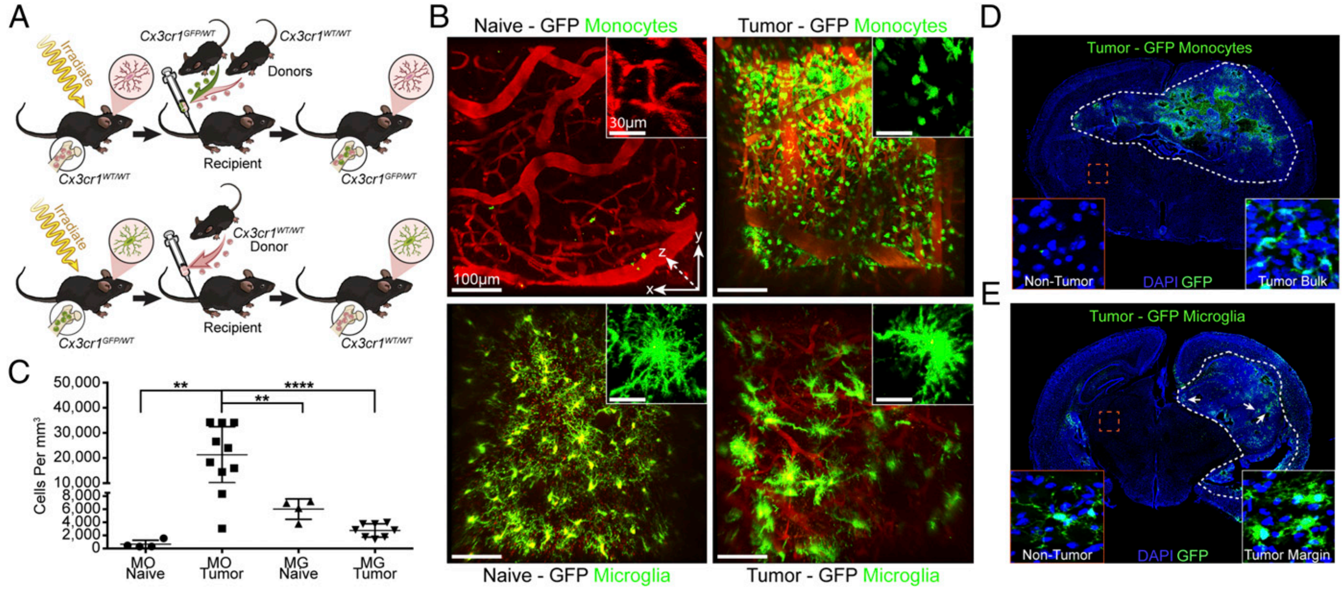

Fig. 2. Reciprocal bone marrow chimera allows for the detection of individual TAM populations. (*A*) Bone marrow chimera schematic depicting the generation of mice with GFP-labeled BMDMs or GFP-labeled microglia. (*B*) Raw Z stack images acquired from 2-photon imaging of naïve and tumor-bearing mice with GFP-BMDMs (*Top*) and GFP-microglia (*Bottom*). (Scale bar: 100 μm; magnified inset scale: 30 μm.) (*C*) GFP-labeled TAM populations were quantified in Imaris, with the number of cells per mm^3 calculated. Each dot represents an individual animal. MO, monocyte; MG, microglia. $^{**}P < 0.01$; $^{****}P < 0.0001$, 1-way ANOVA. (*D*) 10× Z-stitch immunofluorescence image of a tumor-bearing $Cx3cr1^{GFP/WT}\rightarrow Cx3cr1^{WT/WT}$ mouse. Note the presence of GFP signal only in tumor tissue. GFP, monocytes; DAPI, nuclei. (*E*) $Cx3cr1^{WT/WT}\rightarrow Cx3cr1^{GFP/WT}$ chimera in a tumor-bearing mouse. Microglia are unevenly distributed in the tumor bulk and accumulate at the tumor margins in distinct clusters (white arrows). GFP, microglia; DAPI, nuclei.

clusters at the periphery of tumor margins (Fig. 2 *B* and *E* and *SI Appendix*, Figs. S3*B* and S4*A*). As previously demonstrated by our laboratory (5), we did not observe an increase in the microglial population in tumor-bearing brains but did find a significant increase in the BMDM population. We also observed a slight increase in GFP intensity in tumor-associated microglia compared with nontumor microglia (*SI Appendix*, Fig. S3*F*). In addition, we found a decrease in microglial branching with increases in surface area and volume in tumor-bearing mice, depicting the accepted notion of microglia adopting an amoeboid morphology in pathological conditions as they phagocytose dead or damaged cells (*SI Appendix*, Fig. S5) (12, 13, 25, 26).

TAMs Dynamically Alter Their Phenotype in the Tumor Microenvironment. After image acquisition, XYZ stacks were imported into Imaris for creation of 3D representations (Fig. 3*A*). Images acquired via 2-photon microscopy were highly amenable for the detection and quantification of morphology, whereas traditional confocal microscopy was unable to resolve the fine microglial processes due to GFP signal loss and reduced Z stack depth (*SI Appendix*, Fig. S4 *C* and *D*). After image processing and quality control measures, morphology analysis was performed on both BMDMs and microglia (*SI Appendix*, Fig. S6). Microglia were found to be larger in mean surface area (5,516 ± 267.3 μm^2) and volume (4,729 ± 258.5 μm^3) compared with BMDMs (970.5 ± 24.32 μm^2 and 2,154 ± 94.24 μm^3 respectively) (Fig. 3 *B* and *C*). Microglia also had significantly more primary branches per cell (6 vs. 1), causing them to assume a less spherical morphology with numerous terminal points (28 vs. 1) (Fig. 3 *D* and *E*).

Time-lapse images were analyzed to determine migratory differences between the two cell types. BMDMs were found to be migratory, consisting of two phenotypically distinct populations (Fig. 4*A*). There were low–GFP-expressing BMDMs that were highly mobile (0.07826 μm/s), reflecting the phenotype of newly infiltrating inflammatory monocytes (Fig. 4*B*) (5). These cells had an average directional track "straightness" of 0.45 (0, nonstraight; 1, perfectly straight), and were visually observed to migrate through the tissue in a nonspecific fashion. High-GFP-expressing BMDMs were observed to be stationary (0.003478 μm/s), reflecting the phenotype of differentiated macrophages (Fig. 4 *B* and *C* and Movies S1 and S2).

Microglia were found to be stationary; however, their processes were continuously extending and retracting in both tumor and nontumor brains (Fig. 4*D*). These results reinforce previous findings that even in the absence of a stimulus, these cells sense their microenvironment through the continuous extension and retraction of ramified projections (27, 28). Although no instances of microglial cell migration were observed, phagocytic behavior was seen in 1 imaging volume (Movies S3 and S4). The complex branching phenotype coupled with the sensing behavior is likely how these cells detect pathogenic insults in the local microenvironment (27).

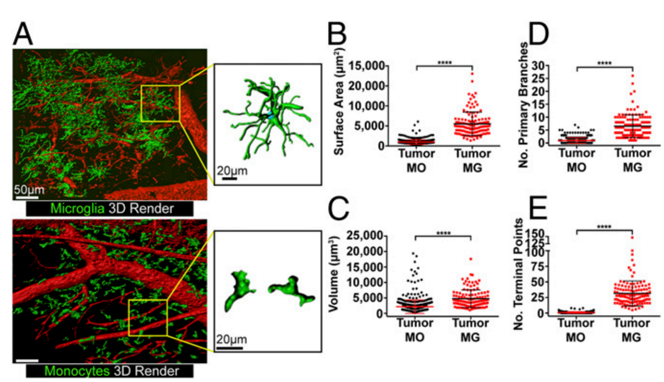

Fig. 3. 3D morphology analysis of TAM populations. (*A*) 3D renderings of GFP-microglia (*Top*) and GFP-BMDMs (*Bottom*) image stacks using Imaris. (Scale bar: 50 μm; magnified inset scale: 20 μm.) (*B*) Single-cell morphology statistics for surface area. $^{****}P < 0.0001$, 2-tailed *t* test. (*C*) Cell volume. $P < 0.0001$, 2-tailed *t* test. (*D*) Number of primary branches for each cell. $P < 0.0001$, Mann–Whitney *U* test. (*E*) Terminal points per cell. $P < 0.0001$, Mann–Whitney *U* test. Tumor MO, 543 cells from 7 mice; tumor MG, 123 cells from 4 mice.

Anti-VEGFA Treatment Reduces Monocyte Infiltration. Given their localization and protumor behavior, TAMs have been suggested to promote resistance to antiangiogenic agents, such as bevacizumab, but the underlying mechanisms remain unknown (29–31). Evidence suggests that resistance may be due to changes in the tumor microenvironment and the presence of myeloid-derived TAMs (32, 33). Mechanisms of TAM infiltration may include passive migration due to a leaky BBB or active migration via chemokine gradients; however, which of these mechanisms is most responsible remains unclear. To address this, we treated mice with a neutralizing anti-VEGFA antibody (B20-4.1.1) and performed 2-photon imaging (Fig. 5*A*). We previously found that B20-4.1.1 reduced tumor vessel area and vascular leakiness in a PDGF-B–driven mouse model (34); therefore, it can be inferred that any reduction observed in the infiltration of TAMs is due in part to normalized vasculature (20, 35, 36). We found increased post-injection survival in tumor-bearing mice treated with B20-4.1.1 compared with vehicle-treated mice (42.5 d vs 30 d) (Fig. 5*B*). In addition, the number of infiltrating monocytes/macrophages was significantly lower in the B20-4.1.1–treated mice, as observed on 2-photon imaging (8,619 cells/mm^3 vs. 21,317 cells/mm^3) (Fig. 5 *C* and *D*). A morphological switch also occurred in the B20-4.1.1–treated mice. Compared with vehicle-treated mice, the cells assumed a more highly branched phenotype (2.46 vs. 0.99 branches per cell) and significantly increased total cell volume (2,661.96 ± 72.82 vs. 2,154 ± 94.24 μm^3) and surface area (1,503 ± 41.78 vs. 970.5 ± 24.41 μm^2), causing their sphericity to decrease (0.668 ± 0.005 vs. 0.734 ± 0.004) (Fig. 5 *E–J*). Interestingly, we did not observe any significant difference in cell motility or cell directionality with B20-4.1.1 treatment (Fig. 5 *K* and *L*). This treatment also appeared to normalize the tumor vasculature; however, due to the leaky nature of our vehicle-treated tumors, the TRITC signal was either too weak or too diffuse to enable a reliable quantitative vascular comparison. These changes suggest that B20-4.1.1 treatment partially prevents the infiltration of circulating BMDMs and induces the differentiation of existing monocytes into tissue macrophages.

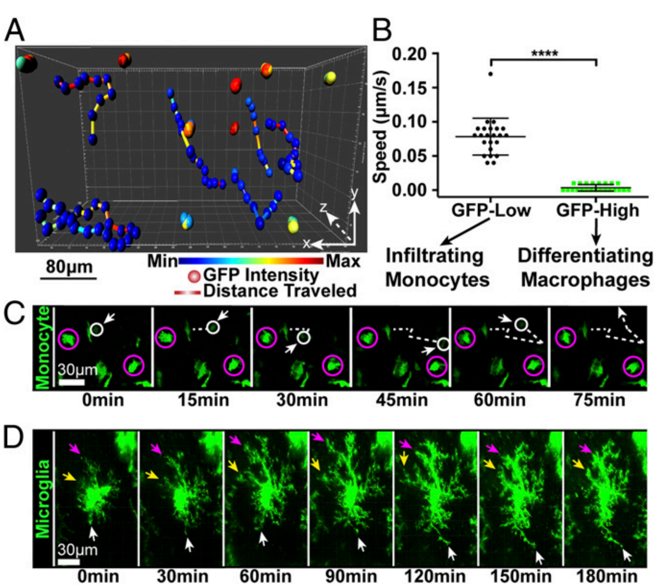

Fig. 4. TAM migration analysis in time-lapse images. (*A*) Cell tracking using the Imaris spots function. Spheres indicate individual cell position at each time point and are colored according to GFP intensity. Movement tracks are colored by distance traveled by each cell. (*B*) Speed of individual cells with low (22 cells from 7 mice) and high (22 cells from 3 mice) GFP intensity. ****$P < 0.0001$, 2-tailed *t* test. (*C*) Individual cell tracking of a low–GFP-expressing monocyte (white circle) compared with high–GFP-expressing cells (purple circles). The dashed line indicates the path of the cell over 75 min. (Scale bar: 30 μm.) (*D*) Time-lapse imaging of a stationary tumor microglia over 180 min. Arrows indicate extension and retraction of individual cell processes. (Scale bar: 30 μm.)

Discussion

Previous morphological studies of microglia in a pathogenic state have been informative, yet future studies may benefit from larger imaging dimensions to reduce the loss of cellular information (37–40). In addition, studies of BMDM morphology in the context of GBM are lacking. The 2D immunofluorescence images acquired from fixed slices 40 μm and thinner inevitably miss the numerous 3D structures of microglia. Our study included 3D image stacks up to 500 μm in vertical depth to ensure that cell bodies and all of their many branches were included in the analyses. Our methodological approach closely resembles recently published findings demonstrating that tumor microglia respond to therapeutic treatment via morphological changes (40); however, contrary to our findings that BMDMs are the predominate TAM subtype in immunocompetent mice, the authors of that study reported greater infiltration of microglia in patient-derived xenograft (PDX) models using immunocompromised nod-scid mice. Further studies are needed to evaluate whether the differences observed in monocyte infiltration in PDX models compared with GEMMs of GBM are a result of the immunocompromised status of nod-scid mice or due to species incompatibility between human chemokines and mouse chemokine receptors (41).

Our findings demonstrate that microglia are large, branched cells with highly active processes that extend and retract in a sensing manner in both naïve and tumor contexts. These results confirm our previous understanding that even in the absence of a stimulus, microglia are constantly surveying their microenvironment through the continuous extension and retraction of ramified projections (27, 28). We did not observe any instances of microglial cell body migration, in line with a previous finding of limited movement of microglial soma (42). Conversely, we found that infiltrating BMDMs are minimally branched and highly migratory. As cells up-regulate their expression of CX3CR1 as indicated by GFP intensity, their migration capacity decreases. These results imply that BMDM cell migration is a function of the state of differentiation or morphological phenotype.

Recent studies have demonstrated that anti-VEGFA therapy causes increased infiltration of monocytic macrophages into the tumor in recurrent GBM using a xenograft mouse model (43). Evidence also suggests that in response to anti-VEGFA blockade, infiltrating myeloid cell populations initially decrease, but then rapidly increase as the tumor progresses (33). Although we found a decrease in this population via 2-photon imaging, our model captured the biology of primary tumors, not recurrent tumors, and was performed in immunocompetent mice, which may explain discrepancies between our findings and those reported by others. Vascular normalization has been attributed to anti-VEGFA therapy, which may explain the early decrease in myeloid cell populations (29, 30, 34); however, as the tumor grows over time, hypoxia increases, providing infiltrating cells with the chemokines and stimulating factors necessary for their recruitment (44–46). In our study, we observed a decrease in the CX3CR1$^+$ myeloid population after B20-4.1.1 treatment. In addition, in response to treatment, monocytes assumed a more branched phenotype, suggesting that VEGFA sequestration promotes differentiation of these cells. As TAMs have been shown to promote tumor vascularization, a possible scenario arises: do BMDMs present during anti-VEGFA therapy become trapped in the tumor and promote vascularization over time (47, 48)? Although not performed in this study, longitudinal imaging over the course of anti-VEGFA treatment could answer this question and may identify mechanisms of resistance that otherwise would go unobserved using other methods.

In conjunction with our previous study demonstrating that genetic loss of CCL2/CCR2 partially decreases monocyte infiltration, our present finding that B20-4.1.1 also partially reduces this population demonstrates that monocyte infiltration is not due

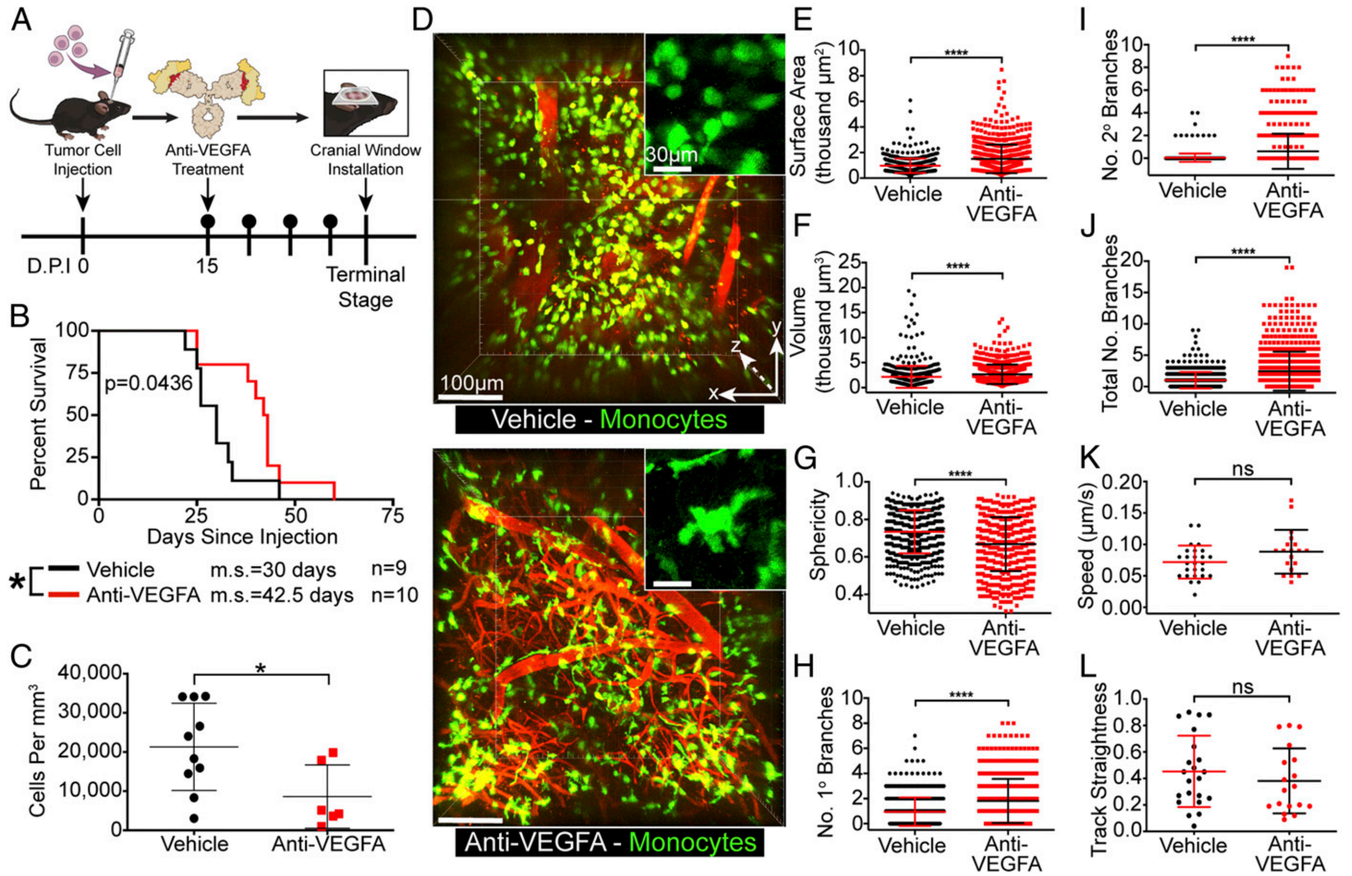

Fig. 5. Anti-VEGFA treatment reduces TAM infiltration and induces a morphological response. (*A*) Timeline of treatment schedule and imaging of anti-VEGFA–treated mice. DPI, days postinjection. (*B*) Kaplan–Meier survival curves for vehicle-treated mice ($n = 9$; median survival, 30 d) and anti-VEGFA–treated mice ($n = 10$; median survival, 42.5 d). $P = 0.0436$. ms, median survival. (*C*) Quantification of cell populations acquired from individual imaging volumes. $^*P < 0.05$. (*D*) Representative image stacks acquired during 2-photon imaging of mice with GFP-labeled BMDMs for vehicle-treated and anti-VEGFA–treated animals. (Scale bar: 100 μm; magnified inset scale: 30 μm.) (*E*) Single-cell morphology statistics for surface area. $^{****}P < 0.0001$, 2-tailed *t* test. (*F*) Cell volume. $P < 0.0001$, 2-tailed *t* test. (*G*) Sphericity (1.0, perfect sphere; 0, not spherical). $P < 0.0001$, 2-tailed *t* test. (*H*) Number of primary branches for each cell. $P < 0.0001$, Mann–Whitney *U* test. (*I*) Number of secondary branches for each cell. $P < 0.0001$, Mann–Whitney *U* test. (*J*) Total number of branches for each cell. $P < 0.0001$, Mann–Whitney *U* test. Vehicle MO, 543 cells from 7 mice; anti-VEGFA MO, 717 cells from 7 mice. (*K*) Speed of individual cells, 2-tailed *t* test. (*L*) Cell migration track straightness (1.0, straight line), 2-tailed *t* test. Vehicle, 26 cells from 5 mice; anti-VEGFA, 19 cells from 5 mice.

solely to a compromised BBB or chemokine gradients (5). Taken together, our data provide a clear rationale for combining anti-VEGF therapy with chemotaxis blockade, such as anti-CCL2/CCR2, to block BMDM infiltration into GBM. The findings of these studies have far-reaching implications beyond their applications in GBM. Open-skull window 2-photon imaging techniques coupled with chimeric mouse models could be applied to the study of stroke and other neurologic disorders in which microglia and BMDMs have been known to play differential roles (49–52). In addition, deciphering mechanisms of macrophage recruitment to pathogenic sites may lead to improved targeted therapies for the treatment of these diseases.

Materials and Methods

Mice. All experimental procedures utilizing live mice were approved by Emory University's Institutional Animal Care and Use Committee (Protocol 2003253). Details are provided in *SI Appendix*.

Bone Marrow Chimera. Bone marrow transplants were performed as reported previously (5) and described in detail in *SI Appendix*.

Orthotopic Cell Injections. We used the RCAS/*tv-a* system to generate de novo primary murine GBM as reported previously (5, 53) and described in detail in *SI Appendix*. Once tumors were formed in donor transgenic mice, tumor cells were freshly dissociated, and 2.5×10^4 cells were superficially injected into the right frontal striatum of recipient mice at 8 to 10 wk after bone marrow transplantation using the following coordinates: bregma AP (anterior/posterior), 0.2 mm; ML (medial/lateral), 1.0 mm, over which a burr hole was made with a sterile 0.7-mm-diameter drill bit (19007-7; Fine Science Tools). A Hamilton syringe affixed to a stereotactic apparatus was then lowered to a depth of 1.0 mm, and 1 μL of cell suspension was injected through a 30-gauge needle.

Skull Window Placement and 2-Photon Imaging. At approximately 3 wk after tumor transplantation, mice began to show neurologic signs of brain tumor. A 5-mm-diameter sterile circular coverslip (Electron Microscopy Sciences) was carefully placed to facilitate visual access to the tumor bulk and tumor margin. At the time of imaging, TRITC-dextran (2.5 mg/mL; Sigma-Aldrich) was injected i.v. to outline the blood vessels. Mice were fixed to a custom-built stage to minimize breathing artifact during image acquisition. Images were acquired with a 25× water immersion objective (Leica; NA 0.95) on a Leica SP8 confocal microscope equipped with a tunable pulsed chameleon infrared multiphoton laser (Coherent) and 2 high-sensitivity hybrid-PMT (HyD) detectors (Leica). High-resolution XYZ stack images (1,024 × 1,024 pixels per Z step) were taken with a step size of 2.5 μm, and images of XYZT stacks were taken every 2.5 min for approximately 3 h on average. At the end of each imaging session, the mice were euthanized and their brains processed for confocal imaging as described previously (5). More details are provided in *SI Appendix*.

B20-4.1.1 Treatment. At 15 d after tumor cell injection, mice were injected i.p. with either anti-VEGFA (B20-4.1.1, 5 mg/kg; Genentech) or vehicle control (bacteriostatic 0.9% sodium chloride) every 4 d until terminal neurologic symptoms appeared.

Image and Data Analysis. High-resolution image stacks were imported into Imaris version 9.0.2 (Bitplane). Total cell numbers within each XYZ stack were quantified using the spots function. Morphology analysis of BMDMs and microglia were done using the surfaces and filaments functions, respectively. Additional details are provided in *SI Appendix*.

Statistical Analysis. Graphs were created using GraphPad Prism 6. Results are expressed as mean ± SD. Comparisons between 2 groups were performed using the unpaired 2-tailed *t* test or nonparametric Mann–Whitney *U* test, as indicated in the figure legends. One-way ANOVA was used to compare the means between 3 or more independent groups. Kaplan–Meier survival analysis was performed using the log-rank test. A P value < 0.05 was considered to indicate statistical significance.

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